

Chapter 5. Commissioning

5.1. Onboarding Payload

The purpose of this section is to define the contents of the Onboarding Payload needed to allow commissioning a device into a Matter network. It also specifies the representation and encoding of said payload as a QR Code, as a manually entered code, and as content in an NFC tag.

5.1.1. Onboarding Payload Contents

The Onboarding Payload is composed of required and optional information which will be used by the Commissioner to ensure interoperability between commissioners and devices and provide a consistent user experience. Some or all of this content will be encoded into different formats, some human-readable (such as numeric string) and machine-readable (such as QR code and NFC Tag) formats for printing or display on or integration into the device. The following are the elements that may be used in an Onboarding Payload for a Matter device.

5.1.1.1. Version

A version indication provides versioning of the payload and SHALL be included. Version for machine-readable formats is 3 bits with an initial version of **0b000**. Version for Manual Pairing Code is 1 bit with an initial version of **0b0**.

Rationale: This allows a way to introduce changes to the payload as needed going into the future.

5.1.1.2. Vendor ID and Product ID

Vendor ID and **Product ID**, each a 16-bit value, SHALL be included in machine-readable formats and MAY be included in the Manual Pairing Code.

Rationale: This allows a way to identify the make and model of the device, which is used further during the commissioning flow, such as during the [Device Attestation procedure](#). These unique identifiers also help to retrieve device model metadata like product name, product description, and firmware update URL from the [Distributed Compliance Ledger](#), as well as information relevant to the commissioning flow (see [Section 5.7, “Device Commissioning Flows”](#)).

5.1.1.3. Custom Flow

A 2-bit unsigned enumeration specifying the [Device Commissioning Flow](#) SHALL be included in machine-readable formats. For the encoding of Custom Flow in the Manual Pairing Code, see [Section 5.1.4.1.2, “Custom Flow for Manual Pairing Code”](#).

Rationale: This guides the Commissioner as to whether steps are needed before commissioning can take place.

- A value of **0** indicates that no steps are needed (apart from powering the device).
- A value of **1** indicates that user interaction with the device (pressing a button, for example) is required before commissioning can take place. The specific steps required can be found in the [CommissioningModeInitialStepsHint](#) field of the [Distributed Compliance Ledger](#) for the given Vendor ID and Product ID.